RMO 2025a

KV or JNV

Exercise 1 (RMO 2025a P1,). Solve the following system of equations in nonnegative integers $a_1, a_2, \ldots, a_8$ where $a_i \neq 1$ for $i = 1, \ldots, 8$:

$$\begin{aligned} a_1a_2 &= a_3 + a_4, \\ a_3a_4 &= a_5 + a_6, \\ a_5a_6 &= a_7 + a_8, \\ a_7a_8 &= a_1 + a_2. \end{aligned}$$

Walkthrough —

(a) Observe that $(a-1)(b-1) = ab - a - b + 1$ holds for any integers a and b.

(b) Using this identity, rewrite the system of equations conveniently.

Solution 1. Let $(a_1, a_2, a_3, a_4, a_5, a_6, a_7, a_8)$ be a solution of the system of equations. Note that

$$\begin{aligned} (a_1-1)(a_2-1) &= a_1a_2 - a_1 - a_2 + 1, \\ (a_3-1)(a_4-1) &= a_3a_4 - a_3 - a_4 + 1, \\ (a_5-1)(a_6-1) &= a_5a_6 - a_5 - a_6 + 1, \\ (a_7-1)(a_8-1) &= a_7a_8 - a_7 - a_8 + 1. \end{aligned}$$

Since $(a_1, a_2, a_3, a_4, a_5, a_6, a_7, a_8)$ satisfies the system of equations, we obtain

$$(a_1-1)(a_2-1) + (a_3-1)(a_4-1) + (a_5-1)(a_6-1) + (a_7-1)(a_8-1) = 4.$$

Since $a_i \neq 1$ for any $i = 1, \ldots, 8$, we have $a_i - 1 \neq 0$ for any $i = 1, \ldots, 8$. Using that $(a_1, a_2, a_3, a_4, a_5, a_6, a_7, a_8)$ satisfies the system of equations, it follows that one of $a_1, \ldots, a_8$ is equal to 0 if and only if all of $a_1, \ldots, a_8$ are equal to 0. Hence, if some a_i is not equal to 0, then all of $a_1, \ldots, a_8$ are greater than or equal to 2, and in this case, we have

$$(a_1-1)(a_2-1) \geq 1,$$

$$(a_3-1)(a_4-1)\geq 1,$$
$$(a_5-1)(a_6-1)\geq 1,$$
$$(a_7-1)(a_8-1)\geq 1,$$

which implies that all of $a_1,\ldots,a_8$ are equal to 2. Also note that $(0,0,0,0,0,0,0,0)$, $(2,2,2,2,2,2,2,2)$ are solutions of the system of equations. This shows that the solutions of the system of equations in nonnegative integers where $a_i\neq 1$ for $i=1,\ldots,8$ are

$$(0,0,0,0,0,0,0,0),\quad (2,2,2,2,2,2,2,2).$$

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Exercise 2 (RMO 2025a P2,). Let a,b,c be positive real numbers satisfying $abc=1$. Prove that

$$\frac{2a^2}{a^3+1}+\frac{2b^2}{b^3+1}+\frac{2c^2}{c^3+1}\leq a^2+b^2+c^2.$$

Walkthrough —

(a)

Solution 2.

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Exercise 3 (RMO 2025a P4,). A frog is initially at $(0,0)$ and it reaches $(n,2)$, $n\geq 1$, using the following moves in any order several times:

1. $R=(1,0)$, that is, if the frog is at (a,b) it goes to $(a+1,b)$;
2. $U=(0,1)$, that is, if the frog is at (a,b) it goes to $(a,b+1)$;
3. $D=(1,1)$, that is, if the frog is at (a,b) it goes to $(a+1,b+1)$.

In how many ways can the frog go from $(0,0)$ to $(n,2)$, $n\geq 1$, using the above steps subject to the condition that steps of the type UU, DD are forbidden? (For example, for $n=3$, $RDUR, DRD$ are admissible paths, while $DDR, RUURR$ are not.)

Walkthrough —

(a)

Solution 3.

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Exercise 4 (RMO 2025a P6,). Define the sequence $\langle a_0, a_1, a_2, \dots \rangle$ as follows: $a_0 = 49$ and $a_n = 10^{2^n} a_{n-1} - 1$ for $n \geq 1$. Show that $s(a_n^2) = n^2 + n + 7$ for all $n \geq 0$, where $s(m)$ denotes the sum of digits in base 10 representation of a nonnegative integer m.

> **Walkthrough** —
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> **(a)**

Solution 4.

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